

Matthew P. Szuromi

Doctoral Student · Theoretical & Computational Neuroscience

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Google Scholar

ORCID

GitHub

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RESEARCH STATEMENT

Seeking to apply deep tools from mathematics and theoretical physics to hard problems in computational and theoretical neuroscience — translating rigorous, first-principles thinking into practical tools and theories.

EDUCATION

2023 – Present	Ph.D., Computational Neuroscience Boston University · Graduate Program in Neuroscience
2016 – 2021	B.S., Pure Mathematics & Physics University of Michigan · College of Literature, Science & the Arts

PUBLICATIONS

Modelling dysfunction-specific interventions for seizure termination in epilepsy

Kamaraj AK & **Szuromi MP** (2025). *npj Systems Biology and Applications*. doi:10.1038/s41540-025-00632-9

Dynamotypes for Dummies: A Toolbox, Atlas, and Tutorial for Simulating a Comprehensive Range of Realistic Synthetic Seizures

Sheckler C, Kish K, Walker Z, Barkelew G, Crisp DN, **Szuromi MP**, Saggio ML & Stacey WC (2025). *eNeuro*, 12(10), ENEURO.0200-25.2025. doi:10.1523/ENEURO.0200-25.2025

Optimization of ictal aborting stimulation using the dynamotype taxonomy

Szuromi MP, Jirsa VK & Stacey WC (2023). *Journal of Computational Neuroscience*, 51, 445–462. doi:10.1007/s10827-023-00859-7

RESEARCH EXPERIENCE

2024 – Present

Ocker Lab — Boston University

Advisor: Dr. Gabriel Ocker

- Building a symbolic–numerical pipeline for automated Feynman-diagram evaluation in nonlinear stochastic differential equations, including a novel proof of a generic loop-diagram counting algorithm
- Developing a statistical field theory for cell-type-specific stochastic spiking neural networks with correlated inputs
- Modelling thalamo-cortical network development with spike-timing-dependent plasticity during critical periods for primary sensory cortices
- Studying stochastic state transitions in spiking networks with spike resets

2023 – Present

BRAIN Dynamics Lab — Johns Hopkins University

Advisors: Dr. Catherine Chu & Dr. Mark Kramer

- Developing a machine-learning approach to detect hippocampal sharp-wave ripples in human stereo-EEG recordings
- Studying the relation between hippocampal sharp-wave ripple activity and memory consolidation in humans

2023 – Present

Independent Research

- Developing low-dimensional neural mass models as practical analytic tools for studying multi-state nonlinear systems, with application to epilepsy and neural dysfunction

2024

Han Lab — Boston University

Advisor: Dr. Xue Han

- Developed an improved action-potential detector for in-vivo voltage-imaging datasets

2020 – 2023

Stacey Lab — University of Michigan

Advisor: Dr. William Stacey

- Developed theoretical understanding of inconsistent efficacy of stimulatory treatments in epilepsy
- Developed a model of seizure dynamics to understand E/I imbalance at seizure onset

2019 – 2020

Levin Lab — University of Michigan

Advisor: Dr. Daniel Levin

- Determined properties of photomultiplier tubes for use in particle detectors

SKILLS

Mathematical

Nonlinear dynamics

Bifurcation theory, timescale separation, coarse-graining

Stochastic dynamics

Computational

Programming

Python, MATLAB, Mathematica, C++, LaTeX, Linux/Unix

Libraries

SDEs, point processes, statistical field theory (path-integral methods)

Graph theory

Graph-enumeration algorithms

Analysis

Real & functional analysis (Volterra-kernel representations of local learning rules)

Statistics & data

Time-series analysis, spectral estimation, GLMs, HMMs

NumPy, SciPy, SymPy, Matplotlib, PyTorch

Packages

NEST, bdtoolbox, MatCont, FOOF, CERN ROOT

PRESENTATIONS & LECTURES

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| 2025 | Controlling and Probing Seizure Dynamotypes
Minisymposium · Computational Neuroscience (CNS 2025) |
| 2023 | Optimization of Ictal Aborting Stimulation
Minisymposium · SIAM Dynamical Systems (DS23) |
| 2022 | Optimization of Ictal Aborting Stimulation
Selected talk · International Conference for Technology and Analysis of Seizures (ICTALS) |
| 2022 | Optimization of Ictal Aborting Stimulation
Invited talk · University of Michigan Neural Networks Journal Club |

TEACHING

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| 2022 | Private Tutor <ul style="list-style-type: none">• Taught Linear Algebra (proofs & applications) one-on-one online; developed practice problems |
| 2019 | Tutor — ACE Tutor Zone, Ann Arbor, MI <ul style="list-style-type: none">• Taught Calculus I/II, elementary mechanics, AP Calculus & Physics, and SAT/ACT Math to individuals and small groups |
| 2018 – 2019 | Undergraduate Tutor — University of Michigan <ul style="list-style-type: none">• Tutored Calculus I/II and elementary mechanics one-on-one |

REFERENCES

Available upon request.